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TGPSC AEE (PCB)

Question Paper Analysis & Detailed Answer Key

Concerned Subject · QP2 · 150 Questions · 300 Marks · No negative marking

31 October 2026 · Saturday · Prepared by DHRION eLearning

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Paper breakup

The paper has 150 questions at 2 marks each (300 marks) in 150 minutes, with no negative marking. It is one section with question shuffling switched on. The subject groups below follow the blocks the question numbers fall into. Every key was checked against standard textbooks. The result:

Result	Question	Count	Marks
Official key correct, nothing to dispute	All others	146	292
Key wrong: challenge	**Q86**	1	2
Key very likely wrong: challenge	**Q85**	1	2
Question flawed: seek deletion or bonus marks	**Q25**	1	2
Key holds, weak alternative: optional	**Q127**	1	2
Total		**150**	**300**

So 4 questions can be challenged. Three are worth filing: Q86 and Q85 have wrong keys, and Q25 should be deleted or given bonus marks. Together they are worth 6 marks. Q127 is a weak case, so raise it only if the commission allows more objections. The full arguments and detailed references are in "Wrong and disputed keys" at the end.

Subject × question format

Subject	Standard MCQ	Statement	Match	A–R	Sequence	Analogy	Correct pair	Total
Engineering Materials Q1–15	7	3	1	0	3	1	0	**15**
Fluid Mechanics & Hydraulics Q16–30	7	4	1	0	2	1	0	**15**
Water, Wastewater, Air & Noise Q31–45	7	4	1	0	2	1	0	**15**
Solid & Agricultural Waste Q46–60	7	4	1	0	2	1	0	**15**
Process Calculations & Thermodynamics Q61–75	7	5	1	0	2	0	0	**15**
Ecosystems Q76–91 + Q117	5	5	1	3	2	0	1	**17**
Natural Resources Q92–106	7	5	1	0	2	0	0	**15**
Biodiversity & Conservation Q107–121, less Q117	8	3	1	0	1	0	1	**14**
Global Issues & Environmental Law Q122–136	7	5	1	0	2	0	0	**15**
EIA, Management & Sustainability Q137–150	7	4	1	0	2	0	0	**14**
Total	**69**	**42**	**10**	**3**	**20**	**4**	**2**	**150**
Share of paper	46.00%	28.00%	6.67%	2.00%	13.33%	2.67%	1.33%	100%

Q117 sits in the biodiversity block but tests ecosystem concepts, so it is counted under Ecosystems.

Q148 is counted as a statement question because it lists pairs and asks for a combination ("1 and 2 only"), the same structure as Q104 and Q134. That leaves only Q89 and Q120 as true "pick the one correct pair" questions.

Questions by format

Format	Count	Question numbers
Two statements, judge true/false	18	8, 9, 10, 26, 27, 42, 43, 54, 55, 70, 84, 100, 101, 115, 116, 129, 130, 144
Multiple statements or pairs, choose the combination	22	28, 29, 44, 45, 56, 57, 68, 69, 71, 73, 86, 87, 99, 102, 104, 117, 118, 131, 132, 134, 146, 148
Two statements, "does II explain I"	2	85, 145
Match the following	10	13, 24, 40, 60, 72, 88, 103, 119, 133, 147
Assertion–Reason (explicit)	3	81, 82, 83
Process sequence	15	14, 15, 23, 39, 41, 53, 58, 90, 91, 105, 106, 135, 136, 149, 150
Ranking by magnitude	5	12, 25, 74, 75, 121
Analogy	4	11, 30, 38, 59
Single correct pair	2	89, 120
Numerical calculation	3	19 (continuity), 61 (first law), 67 (partial molar volume)

The numericals are also counted as standard MCQs. In total, 81 questions (54%) use a format other than a plain single-answer MCQ, but most test basic concepts. Q135 and Q136 are the same question asked twice.

Options are shuffled in the live test, so learn the answer text, not the option number. References are given for the four questions that can be challenged; the standard textbooks for each subject are listed at the end.

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Q1–15 · Engineering Materials

Q1 · Standard MCQ

Porosity of a material is expressed as the ratio of the volume of its pores to the:

Options:

1. Volume of the specimen
2. Mass of the specimen
3. Density of water
4. Volume of the solid particles alone

Official key: (1) Volume of the specimen · **Verdict:** Correct

Porosity = volume of voids ÷ total (bulk) volume of the specimen, usually as a percentage. Voids ÷ solids is the void ratio, not porosity.

Q2 · Standard MCQ

Which mechanical property of a material is defined as its ability to resist penetration by a harder body?

Options:

1. Hardness
2. Elasticity
3. Plasticity
4. Toughness

Official key: (1) Hardness · **Verdict:** Correct

Hardness is resistance to indentation or penetration, measured by the Brinell, Vickers and Rockwell tests. Toughness is the energy absorbed before fracture.

Q3 · Standard MCQ

The ratio of stress to strain within the proportional limit of a material is known as its:

Options:

1. Modulus of elasticity
2. Percentage elongation
3. Poisson's ratio
4. Ultimate tensile strength

Official key: (1) Modulus of elasticity · **Verdict:** Correct

By Hooke's law, $\sigma = E\varepsilon$ within the proportional limit, so stress ÷ strain = E, Young's modulus.

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Q4 · Standard MCQ

When a mild steel specimen undergoes tensile testing, the percentage elongation measured after fracture is primarily used to assess which mechanical property of the material?

Options:

1. Ductility
2. Hardness
3. Density
4. Toughness

Official key: (1) Ductility · **Verdict:** Correct

Percentage elongation, (Lf − L0)/L0 × 100, and percentage reduction in area are the standard measures of ductility.

Q5 · Standard MCQ

Iron exists in different allotropic forms; the crystal structure of gamma iron is:

Options:

1. Face-centred cubic structure
2. Body-centred cubic structure
3. Hexagonal close-packed structure
4. An amorphous, non-crystalline structure

Official key: (1) Face-centred cubic structure · **Verdict:** Correct

α-iron is BCC below 912 °C, γ-iron (austenite) is FCC from 912 to 1394 °C, and δ-iron is BCC from 1394 to 1538 °C.

Q6 · Standard MCQ

Gamma iron (austenite), with its face-centred cubic structure, is stable over approximately which temperature range?

Options:

1. 912 °C to 1394 °C
2. 0 °C to 500 °C
3. 1394 °C to 1536 °C
4. Below 273 °C only

Official key: (1) 912 °C to 1394 °C · **Verdict:** Correct

These are the allotropic transformation temperatures of pure iron. Option 3 is the δ-ferrite (BCC) range.

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Q7 · Standard MCQ

Normalizing of steel consists of heating it above the critical range followed by:

Options:

1. Rapid quenching in water
2. Quenching in oil
3. Cooling slowly inside the furnace
4. Cooling in still air

Official key: (4) Cooling in still air · **Verdict: Correct**

Normalizing means heating about 40–50 °C above A_3 or Acm, soaking, then cooling in still air. Furnace cooling is annealing; quenching is hardening.

Q8 · Two statements

Refer to the two statements given below about the elastic behaviour of engineering materials, and choose the CORRECT option.

Statement A: In the linear elastic region of an engineering stress-strain curve, the slope represents the modulus of elasticity.

Statement B: Under the same applied stress within the linear elastic region, a material with a higher modulus of elasticity undergoes greater elastic strain.

Options:

1. Statement A is true, but Statement B is false.
2. Statement B is true, but Statement A is false.
3. Both Statements A and B are true.
4. Both Statements A and B are false.

Official key: (1) A true, B false · **Verdict: Correct**

The slope of the elastic line is E. Since $\varepsilon = \sigma/E$, a higher E gives a smaller strain for the same stress.

Q9 · Two statements

Consider the following statements regarding the stress-strain behaviour of mild steel and choose the CORRECT option.

Statement A: After the yield region, mild steel undergoes strain hardening, during which stress increases with further increase in strain.

Statement B: During strain hardening, stress again becomes directly proportional to strain and remains so up to the ultimate tensile stress.

Options:

1. Statement A is true, but Statement B is false.
2. Statement B is true, but Statement A is false.
3. Both Statements A and B are true.
4. Both Statements A and B are false.

Official key: (1) A true, B false · **Verdict: Correct**

Strain hardening is plastic and non-linear. Proportionality holds only below the proportional limit.

Q10 · Two statements

Consider the following statements regarding the determination of phase compositions in a two-phase region of a binary equilibrium phase diagram, and choose the CORRECT option.

Statement A: At a specified temperature within a two-phase region, the composition of each equilibrium phase is determined from the intersection of a horizontal tie line with the corresponding phase boundary.

Statement B: At a specified temperature within a two-phase region, the overall alloy composition alone is sufficient to determine the composition of each equilibrium phase.

Options:

1. Statement A is true, but Statement B is false.
2. Statement B is true, but Statement A is false.
3. Both Statements A and B are true.
4. Both Statements A and B are false.

Official key: (1) A true, B false · **Verdict:** Correct

The ends of the tie line give the phase compositions. The overall composition fixes only the phase amounts, through the lever rule.

Q11 · Analogy

Select the option that shows a process-effect relationship analogous to the pair given below.

Annealing : Relief of internal stresses

Options:

1. Normalizing : Refinement of grain structure
2. Hardening : Increase in ductility
3. Tempering : Increase in brittleness
4. Quenching : Improvement in machinability

Official key: (1) Normalizing : Refinement of grain structure · **Verdict:** Correct

Normalizing refines grains. Each of the other options reverses the true effect: hardening lowers ductility, tempering reduces brittleness, and quenching hurts machinability.

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Q12 · Ranking

Arrange the following cubic crystal structures in decreasing order of the effective number of atoms contained in one conventional unit cell and choose the CORRECT option.

P. Simple cubic (SC)

Q. Face-centred cubic (FCC)

R. Body-centred cubic (BCC)

Options:

1. R > Q > P
2. P > R > Q
3. Q > P > R
4. Q > R > P

Official key: (4) Q > R > P · **Verdict:** Correct

FCC has $8 \times 1/8 + 6 \times 1/2 = 4$ atoms per cell, BCC has $8 \times 1/8 + 1 = 2$, and SC has $8 \times 1/8 = 1$.

Q13 · Match the following

Match the successive stages of tensile behaviour of mild steel in Column I with the corresponding description in Column II and select the CORRECT option.

Stage of tensile behaviour	Description of stress–strain behaviour
P. Initial linear stage	1. Engineering stress decreases after the ultimate tensile point until fracture.
Q. Yield stage	2. Stress is proportional to strain
R. Post-yield hardening stage	3. Large strain occurs with little change in stress
S. Post-ultimate stage	4. Stress increases with plastic strain during strain hardening.

Options:

1. P-2, Q-4, R-3, S-1
2. P-3, Q-2, R-4, S-1
3. P-2, Q-3, R-1, S-4
4. P-2, Q-3, R-4, S-1

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Official key: (4) P-2, Q-3, R-4, S-1 · **Verdict:** Correct

The stages are linear elastic, then yield plateau, then strain hardening, then necking, where engineering stress falls.

Q14 · Sequence

Arrange the following steps in the correct sequence for determining the equilibrium compositions of the two phases present at a specified temperature in a two-phase region of a binary equilibrium phase diagram, and choose the CORRECT option.

P. Project the tie-line intersection points onto the composition axis.

Q. Locate the alloy using its specified temperature and overall composition.

R. Read the equilibrium composition of each phase from the composition axis.

S. Draw a horizontal tie line through the specified temperature to the boundaries of the two-phase region.

Options:

1. Q → P → S → R
2. S → Q → P → R
3. Q → S → R → P
4. Q → S → P → R

Official key: (4) Q → S → P → R · **Verdict:** Correct

Locate the point, draw the tie line, drop perpendiculars to the axis, then read the compositions.

Q15 · Sequence

Arrange the following operations in the correct chronological order for an annealing heat-treatment cycle, and choose the CORRECT option.

P. Cool the metal at the predetermined rate.

Q. Heat the metal to the predetermined temperature.

R. Obtain the desired structure and properties after cooling.

S. Soak the metal until its structure becomes uniform throughout the mass.

Options:

1. S → Q → P → R
2. Q → P → S → R
3. Q → S → P → R
4. Q → S → R → P

Official key: (3) Q → S → P → R · **Verdict:** Correct

Any heat-treatment cycle runs heat, soak, cool, then the resulting structure.

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Q16–30 · Fluid Mechanics & Hydraulics

Q16 · Standard MCQ

Which of the following describes the compressibility of air when subjected to pressure changes?

Options:

1. Air density remains nearly constant with pressure changes
2. Air volume remains nearly constant under pressure changes
3. Air density changes appreciably with pressure changes
4. Air behaves nearly like an incompressible fluid

Official key: (3) Air density changes appreciably with pressure changes · **Verdict:** Correct

For a gas, $\rho = P/RT$, so density varies with pressure. Liquids are nearly incompressible; water's bulk modulus is about 2.2 GPa.

Q17 · Standard MCQ

When pressure is applied at a point on a confined fluid at rest, how is the pressure transmitted according to Pascal's law?

Options:

1. It remains restricted to the point of application.
2. It is transmitted equally in all directions.
3. It is transmitted only along the chamber boundaries.
4. It is transmitted only upward from the point of application.

Official key: (2) It is transmitted equally in all directions · **Verdict:** Correct

Pascal's law states that the intensity of pressure at a point in a static fluid is equal in all directions, and an applied pressure is transmitted undiminished. This is the principle of the hydraulic press.

Q18 · Standard MCQ

In fluid kinematics, the motion of a fluid can be described either by following individual particles or by observing the flow properties at fixed locations. Which statement correctly distinguishes these two approaches?

Options:

1. The Lagrangian method observes fixed locations, while the Eulerian method tracks fluid particles
2. The Lagrangian method tracks fluid particles, while the Eulerian method observes fixed locations
3. The Lagrangian method tracks temperature changes, while the Eulerian method tracks pressure changes
4. The Lagrangian method describes steady flow, while the Eulerian method describes unsteady flow

Official key: (2) · **Verdict:** Correct

The Lagrangian method follows a particle. The Eulerian method records properties at fixed points in space over time.

Q19 · Numerical

Water flows steadily through a pipe of varying diameter. The diameters at sections 1 and 2 are 100 mm and 200 mm, respectively. If the velocity at section 1 is 4 m/s, what are the discharge and velocity at section 2?

Options:

1. 0.0314 m^3/s, 1 m/s
2. 0.0314 m^3/s, 2 m/s
3. 0.0628 m^3/s, 1 m/s
4. 0.0628 m^3/s, 2 m/s

Official key: (1) 0.0314 m^3/s, 1 m/s · **Verdict:** Correct

$$Q = \frac{\pi}{4}(0.1)^2 \times 4 = 0.0314 \text{ m}^3/\text{s}$$

$$V_2 = V_1\left(\frac{D_1}{D_2}\right)^2 = 4 \times \frac{1}{4} = 1 \text{ m/s}$$

Q20 · Standard MCQ

In the graphical representation of Bernoulli's equation, which line represents the sum of pressure head, velocity head, and datum head and remains constant when there is no energy loss along the flow?

Options:

1. The total head line
2. The velocity head line
3. The piezometric head line
4. The datum line

Official key: (1) The total head line · **Verdict:** Correct

The total energy line is $p/\rho g + V^2/2g + z$ and is horizontal with no losses. The piezometric line, or HGL, is $p/\rho g + z$.

Q21 · Standard MCQ

For steady, incompressible flow through a Venturi meter, what is the effect of a smaller throat cross-sectional area on the fluid velocity at the throat compared with the inlet velocity?

Options:

1. It increases the fluid velocity at the throat due to continuity.
2. It decreases the velocity at the throat compared to the inlet.
3. It leaves the fluid velocity at the throat unchanged.
4. It causes a pressure drop only at the inlet section of the meter.

Official key: (1) · **Verdict:** Correct

Since $A_1V_1 = A_2V_2$, a smaller area gives a higher velocity, and Bernoulli's equation then gives a lower pressure at the throat.

Q22 · Standard MCQ

Which combination of fundamental equations is used to derive the head-loss relation for a sudden enlargement in a pipe?

Options:

1. Momentum, energy, and continuity equations
2. Darcy–Weisbach, friction, and uniform-flow equations
3. Continuity, hydrostatic-pressure, and uniform-flow equations
4. Bernoulli, hydrostatic-pressure, and pipe-friction equations

Official key: (1) Momentum, energy, and continuity equations · **Verdict:** Correct

The result, $h_L = (V_1 - V_2)^2/2g$, comes from the momentum equation on the control volume, Bernoulli's (energy) equation, and continuity.

Q23 · Sequence

Which sequence correctly describes the steps involved in determining the bulk modulus of a gas under isothermal conditions?

Options:

1. Calculate bulk modulus, apply pressure, maintain constant temperature, measure volume change.
2. Measure volume change, apply pressure, calculate bulk modulus, maintain constant temperature.
3. Calculate bulk modulus, maintain constant temperature, apply pressure, measure volume change.
4. Maintain constant temperature, apply pressure, measure volume change, calculate bulk modulus.

Official key: (4) · **Verdict:** Correct

$K = -dp/(dV/V)$. Hold T constant, apply dp, measure dV, then compute. For an isothermal gas, $K = p$.

Q24 · Match the following

Match each flow description in Column A with its correct flow type in Column B.

Column A	Column B
1. Flow properties at a fixed point do not change with time	A. Uniform flow
2. Flow properties at a fixed point change with time	B. Non-uniform flow
3. Velocity is identical from one location to another at a given instant	C. Steady flow
4. Velocity differs from one location to another at a given instant	D. Unsteady flow

Options:

1. 1-A, 2-B, 3-C, 4-D
2. 1-D, 2-C, 3-B, 4-A
3. 1-B, 2-A, 3-D, 4-C
4. 1-C, 2-D, 3-A, 4-B

Official key: (4) 1-C, 2-D, 3-A, 4-B · **Verdict:** Correct

Steady and unsteady describe change with time at a point; uniform and non-uniform describe change with space at an instant.

Q25 · Ranking

Given four points along a streamline, each with the following heads: A (20 m), B (25 m), C (15 m), D (30 m). Arrange them in order of decreasing total head as per Bernoulli's principle.

Options:

1. C, A, B, D
2. A, C, B, D
3. D, B, A, C
4. B, D, C, A

Official key: (3) D, B, A, C · **Verdict:** Key matches the numbers, but the question is flawed (see "Wrong and disputed keys")

Ranked numerically, 30 > 25 > 20 > 15 gives D, B, A, C. Under Bernoulli's principle, however, total head is constant along one streamline in ideal flow. With losses it can only fall in the flow direction. Four different "total heads" on one streamline therefore contradict the principle the question cites. In the exam, mark (3).

Reference: Bansal, chapter on dynamics of fluid flow (Bernoulli's equation and its assumptions).

Q26 · Two statements

Consider the following regarding Buckingham's Pi theorem and select the correct option.

(1) It aids in forming dimensionless parameters.

(2) It is not applicable to fluid mechanics problems.

Options:

1. Both statements are correct
2. Only statement 1 is correct
3. Only statement 2 is correct
4. Neither statement is correct

Official key: (2) Only statement 1 is correct · **Verdict:** Correct

The theorem says n variables in m fundamental dimensions form (n − m) π-terms. It is a core tool in fluid mechanics.

Q27 · Two statements

Consider the following statements regarding a steady boundary layer developing over a smooth flat plate under zero-pressure-gradient conditions:

Statement I: The boundary layer thickness increases with distance from the leading edge.

Statement II: The velocity gradient at the wall generally decreases with distance from the leading edge.

Which of the above statements is/are correct?

Options:

1. Neither Statement I nor Statement II is correct
2. Only Statement II is correct
3. Only Statement I is correct
4. Both Statement I and Statement II are correct

Official key: (4) Both correct · **Verdict: Correct**

In laminar flow, $\delta \approx 5x/\sqrt{Re_x}$, which grows as $\sqrt{x}$. Wall shear τw is proportional to $x^{-1/2}$, so the wall velocity gradient falls with x.

Q28 · Multiple statements

Examine the following statements about the limitations and application of Froude model law:

1. It is not suitable for flows where viscous effects dominate.
2. It is widely used for modelling free-surface flows such as those over spillways, weirs and in open channels.
3. It can be used for compressible fluid flows without modification.
4. It requires geometric similarity between model and prototype.

Which of these statements are correct?

Options:

1. 2 and 3 only
2. 1, 3 and 4 only
3. 1, 2 and 4 only
4. 2, 3 and 4 only

Official key: (3) 1, 2 and 4 only · **Verdict: Correct**

Froude's law is for gravity-dominated, free-surface flows. Viscous-dominated flows use Reynolds' law and compressible flows use Mach's law. Geometric similarity is always required.

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Q29 · Multiple statements

Which of the following statements about boundary layer growth over a flat plate are correct?

1. The boundary layer thickness increases with distance from the leading edge.
2. The fluid velocity at the plate surface is zero due to the no-slip condition.
3. The boundary layer remains laminar at all distances from the leading edge.
4. The boundary layer separates from the plate at zero distance from the leading edge.

Options:

1. 1 and 3 only
2. 1 and 2 only
3. 2 and 4 only
4. 3 and 4 only

Official key: (2) 1 and 2 only · **Verdict:** Correct

Transition to turbulence occurs near $Re_x \approx 5 \times 10^5$, so statement 3 is false. Separation needs an adverse pressure gradient and does not occur at the leading edge, so statement 4 is false.

Q30 · Analogy

Select the pair in which the first term is used to determine the second term, similar to the relationship shown in the given pair.

Darcy-Weisbach equation : Major loss

1. Bernoulli's equation : Loss coefficient
2. Hazen-Williams equation : Minor loss
3. Moody chart : Friction factor
4. Reynolds number : Wall roughness

Official key: (3) Moody chart : Friction factor · **Verdict:** Correct

The Moody chart gives f from Re and ε/D, just as $h_f = fLV^2/2gD$ gives the major loss. Hazen–Williams gives friction (major) loss, not minor loss.

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Q31–45 · Water, Wastewater, Air & Noise Pollution

Q31 · Standard MCQ

What is the recommended maximum acceptable limit for chloride in drinking water, according to IS 10500:2012, in mg/L?

Options:

1. 100
2. 500
3. 250
4. 45

Official key: (3) 250 · **Verdict:** Correct

Chloride has an acceptable limit of 250 mg/L and a permissible limit, with no alternative source, of 1000 mg/L. 45 mg/L is the nitrate limit.

Q32 · Standard MCQ

Which parameter is used in IS 10500:2012 to indicate possible bacterial contamination in drinking water?

Options:

1. Hardness
2. Total Coliforms
3. Calcium
4. Iron

Official key: (2) Total Coliforms · **Verdict:** Correct

Total coliforms and *E. coli* shall not be detectable in any 100 mL sample.

Q33 · Standard MCQ

Which layer in a slow sand filter is primarily responsible for biological treatment of water?

Options:

1. Supporting stone layer
2. Gravel layer
3. Coarse sand bed layer
4. Schmutzdecke layer

Official key: (4) Schmutzdecke layer · **Verdict:** Correct

The schmutzdecke is the biological film on top of the sand, made of algae, bacteria and protozoa. It removes pathogens and organic matter.

Q34 · Standard MCQ

Which disinfectant is most suitable for maintaining a residual in drinking water distribution systems to provide continued protection against microbial contamination?

Options:

1. Phenyl
2. Chlorine
3. Ozone
4. UV radiation

Official key: (2) Chlorine · **Verdict:** Correct

Only chlorine leaves a lasting residual; at least 0.2 mg/L of free chlorine is required at the consumer end. Ozone and UV leave none.

Q35 · Standard MCQ

Which characteristic is most important when selecting a pump for handling domestic wastewater in the primary treatment stage?

Options:

1. Ability to develop very high discharge pressure
2. Suitability only for clear, solids-free water
3. Ability to handle solids in suspension
4. Low temperature resistance

Official key: (3) Ability to handle solids in suspension · **Verdict:** Correct

Sewage carries rags and grit, so non-clog pumps with open impellers are used.

Q36 · Standard MCQ

What is the primary purpose of a secondary clarifier in a domestic wastewater treatment plant?

Options:

1. To remove grit and sand from treated effluent
2. To add chemicals for precipitation
3. To disinfect the final effluent
4. To separate biological solids from treated effluent

Official key: (4) · **Verdict:** Correct

The secondary clarifier settles biomass after the biological reactor, and returned sludge (RAS) goes back to the aeration tank.

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Q37 · Standard MCQ

Which type of sludge thickener uses a slow-moving rake mechanism to enhance solids settling?

Options:

1. Belt filter press
2. Centrifuge thickener
3. Gravity thickener
4. Dissolved air flotation thickener

Official key: (3) Gravity thickener · **Verdict:** Correct

In a gravity thickener, a slowly rotating rake or picket fence releases trapped water and moves sludge to the hopper.

Q38 · Analogy

A pair of terms related to air pollution sources and their impact is given below, followed by four alternatives. Select the correct alternative that shows the same relationship as the given pair.

Rural areas : Agricultural runoff :: Urban areas : ?

Options:

1. Burning of coal
2. Vehicular emissions
3. Soil erosion
4. Crop residue burning

Official key: (2) Vehicular emissions · **Verdict:** Correct (stem has a minor flaw)

The pattern is area : its characteristic pollution source, and vehicles dominate urban pollution. Agricultural runoff is really a water-pollution source, although the stem says “air pollution”. That flaw does not change the answer.

Q39 · Sequence

Arrange the following sequential stages of lake eutrophication in CHRONOLOGICAL (increasing time of progression) order, starting from the discharge of nutrient-rich wastewater into a freshwater lake:

1. Rapid proliferation and algal blooms over the lake surface
2. Death and microbial decomposition of dense biomass by aerobic decomposers
3. Severe depletion of dissolved oxygen leading to fish kills and septic conditions
4. Influx and enrichment of limiting nutrients (nitrogen and phosphorus)

Options:

1. 4 → 1 → 2 → 3
2. 1 → 4 → 3 → 2
3. 4 → 3 → 1 → 2
4. 2 → 1 → 4 → 3

Official key: (1) 4 → 1 → 2 → 3 · **Verdict:** Correct

The order is N and P enrichment, algal bloom, die-off and aerobic decay, then DO depletion and fish kills.

Q40 · Match the following

Match the noise monitoring instrument or technique in Column I with its primary application in Column II.

Column I (Instrument or technique)	Column II (Primary application)
A. Sound Level Meter with Data Logger	I. Evaluates individual worker cumulative noise dose over a work shift
B. Personal Noise Dosimeter	II. Provides unattended, continuous recording of ambient urban sound levels
C. Handheld Sound Meter without Logger	III. Identifies predominant frequency components of complex machinery noise
D. Octave Band Analyzer	IV. Performs rapid spot checks and instantaneous preliminary assessments

Options:

1. A-I, B-II, C-IV, D-III
2. A-II, B-IV, C-I, D-III
3. A-III, B-I, C-IV, D-II
4. A-II, B-I, C-IV, D-III

Official key: (4) A-II, B-I, C-IV, D-III · **Verdict:** Correct

A logger records continuously, a dosimeter measures a worker's shift dose, a handheld meter does spot checks, and an octave-band analyser gives the frequency spectrum.

Q41 · Sequence

Arrange the following stages of municipal and industrial waste remediation management from soil pollution in the correct chronological order of systematic implementation from earliest preventative action to ultimate resolution:

A. Bioremediation and vegetative stabilization of contaminated substrate

B. Source reduction and implementation of statutory disposal regulations

C. Site containment, characterization, and hazardous waste isolation

D. Long-term post-remedial environmental monitoring and soil quality testing

Options:

1. C → A → B → D
2. B → A → C → D
3. B → C → A → D
4. A → C → B → D

Official key: (3) B → C → A → D · **Verdict:** Correct

The order is prevention, then containment and characterisation, then remediation, then monitoring.

Q42 · Two statements

Refer to the two statements given below about unit operations and tank configurations in water treatment. Choose the CORRECT option.

Statement A: Continuous-flow sedimentation basins, such as rectangular and circular settling tanks, are designed to retain water at low velocities to allow suspended particles to settle under gravity.

Statement B: Aeration tanks are primarily designed to settle suspended particulate matter by maintaining calm, stagnant hydraulic conditions.

Options:

1. Statement A is true, but Statement B is false
2. Statement B is true, but Statement A is false
3. Both Statements A and B are true
4. Both Statements A and B are false

Official key: (1) A true, B false · **Verdict:** Correct

Aeration tanks are deliberately turbulent so that they transfer oxygen and keep biomass in suspension. They are not settling units.

Q43 · Two statements

Refer to the two statements given below about the biological and physical processes occurring inside a septic tank. Choose the CORRECT option.

Statement A: Inside a septic tank, settled organic matter and retained solids are decomposed by anaerobic bacteria into simpler organic compounds, gases, and mineral matter.

Statement B: A septic tank primarily operates as an aerobic treatment unit where forced aeration removes dissolved organic carbon under oxygen-rich conditions.

Options:

1. Statement A is true, but Statement B is false
2. Statement B is true, but Statement A is false
3. Both Statements A and B are true
4. Both Statements A and B are false

Official key: (1) A true, B false · **Verdict:** Correct

A septic tank is an anaerobic settling-and-digestion unit with no aeration.

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Q44 · Multiple statements

Choose the correct statements regarding the parameters and criteria used to determine compliance with National Ambient Air Quality Standards (NAAQS) for particulate matter in India.

A. Compliance for PM10 is evaluated against both an annual arithmetic mean and a 24-hour average concentration limit.

B. The 24-hour ambient standard limit for PM10 in residential and industrial areas is 100 micrograms per cubic meter.

C. Regulatory compliance requires that ambient particulate matter standards are monitored on a 1-hour maximum peak interval.

D. The annual standard specifies an arithmetic mean of minimum 104 measurements in a year taken twice a week 24-hourly at uniform intervals.

Options:

1. Only A and C
2. A, B, and D
3. B, C, and D
4. Only B and C

Official key: (2) A, B and D · **Verdict:** Correct

PM10 has limits of 60 µg/m³ annual and 100 µg/m³ over 24 hours. Note 1 of the notification gives the 104-measurement rule. There is no 1-hour PM standard, so C is false.

Q45 · Multiple statements

Choose the correct statements regarding effective engineering and policy strategies for mitigating urban air pollution from vehicular emissions.

A. Implementing stringent fuel quality standards, such as lowering fuel sulfur content, substantially reduces toxic exhaust and secondary particulate formation.

B. Promoting organized carpooling and rapid public transit networks decreases personal vehicle density and reduces overall fleet emissions.

C. Expanding urban road widths without traffic demand management indefinitely eliminates tailpipe emission intensity by permanently preventing vehicular congestion.

D. Installing onboard catalytic converters and enforcing emission standards under periodic vehicular inspection programs directly curbs tailpipe carbon monoxide and unburnt hydrocarbon releases.

Options:

1. A, B, and D
2. Only B and C
3. Only A and C
4. B, C, and D

Official key: (1) A, B and D · **Verdict:** Correct

Road widening without demand management induces more traffic, so C is false.

Q46–60 · Solid & Agricultural Waste

Q46 · Standard MCQ

What is the purpose of using trailers in municipal solid waste management?

Options:

1. They are mainly used for street sweeping operations
2. They allow efficient bulk waste transfer between facilities
3. They minimize manual sorting of collected waste
4. They directly compact waste at collection points

Official key: (2) · **Verdict:** Correct

Large trailers loaded at transfer stations haul waste in bulk over long distances, which cuts the cost per tonne-km.

Q47 · Standard MCQ

Which of the following best defines segregation in municipal solid waste management?

Options:

1. Mixing of solid wastes at the transfer station
2. Compressing solid waste for transportation
3. Separating solid waste into different categories
4. Burning of municipal waste at disposal site

Official key: (3) · **Verdict:** Correct

Segregation means sorting waste into streams such as wet, dry and domestic hazardous, ideally at source.

Q48 · Standard MCQ

In the case of a solid waste disposal by incineration process, which of the following best describes the waste movement in a fluidized bed incinerator?

Options:

1. Waste moves downward through successive hearths
2. Waste travels along an inclined stationary grate
3. Waste moves by rotation in a horizontal cylinder
4. Waste is kept suspended by upward air flow

Official key: (4) · **Verdict:** Correct

Upward air fluidises a hot sand bed with the waste in it. Option 1 describes a multiple-hearth incinerator, and option 3 a rotary kiln.

Q49 · Standard MCQ

Which of the following features distinguishes the ramp method from the area method of landfilling?

Options:

1. Application of soil cover after waste placement
2. Use of flat ground for waste placement
3. Use of sloped ground for waste placement
4. Compaction of waste after spreading waste

Official key: (3) · **Verdict:** Correct

The ramp, or slope, method places waste on a sloping face and takes cover soil from just ahead. The area method uses flat ground.

Q50 · Standard MCQ

Which of the following heavy metals commonly associated with e-waste is particularly known for causing serious neurological and neurodevelopmental effects upon exposure?

Options:

1. Beryllium
2. Lead
3. Nickel
4. Chromium

Official key: (2) Lead · **Verdict:** Correct

Lead from CRT glass and solder is a neurotoxin that is especially harmful to children's development.

Q51 · Standard MCQ

When evaluating the chemical characteristics of solid waste, which of the following analytical approaches is specifically used to assess the biodegradable organic load in the liquid runoff or aqueous extract of a solid waste sample?

Options:

1. Biochemical oxygen demand analysis
2. Proximate analysis of waste sample
3. pH measurement of waste extract
4. Heavy metal extraction test

Official key: (1) BOD · **Verdict:** Correct

BOD measures the biodegradable organic load. Proximate analysis (moisture, volatile matter, fixed carbon, ash) is used to judge energy recovery.

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Q52 · Standard MCQ

Which of the following is the most appropriate method for managing large-scale agricultural waste with minimal environmental impact in rural India?

Options:

1. Dumping the waste in municipal landfills
2. Burning the waste in open fields
3. Discharging the waste into rivers
4. Composting the waste on-site

Official key: (4) Composting on-site · **Verdict:** Correct

On-site composting returns nutrients to the soil and avoids burning and transport.

Q53 · Sequence

In a waste collection and transport system where a container is lifted or towed to a processing or disposal facility, arrange the following operations in the correct sequence, assuming hydraulic tipping is used to unload the waste at the destination.

1. Hydraulic tipping of the trailer
2. Towing/lifting of the waste container
3. Transfer of waste towards the processing/disposal facility

Options:

1. 2–3–1
2. 1–2–3
3. 3–2–1
4. 2–1–3

Official key: (1) 2–3–1 · **Verdict:** Correct

In a hauled-container system, the container is picked up, hauled to the facility, then tipped.

Q54 · Two statements

Evaluate the following statements concerning the pathogen inactivation and sterilisation objectives of municipal solid waste incineration, and select the correct option.

Statement I: Incineration under appropriate conditions can ensure destruction of pathogens in municipal solid waste.

Statement II: Sterilisation of waste is one of the objectives associated with incineration.

Options:

1. Both Statement I and Statement II are correct
2. Both Statement I and Statement II are incorrect
3. Statement I is correct, but Statement II is incorrect
4. Statement I is incorrect, but Statement II is correct

Official key: (1) Both correct · **Verdict:** Correct

Incineration at 850 °C or more destroys pathogens. Its objectives include volume reduction, sterilisation and energy recovery.

Q55 · Two statements

Evaluate the following statements regarding the sources of landfill gas generation in municipal solid waste and select the correct answer based on their technical accuracy.

Statement I: Landfill gas is generated from the biodegradable fraction of municipal solid waste.

Statement II: Glass, plastics and metals are major sources of landfill gas generation.

Options:

1. Both Statement I and Statement II are correct
2. Statement I is correct, but Statement II is incorrect
3. Statement I is incorrect, but Statement II is correct
4. Both Statement I and Statement II are incorrect

Official key: (2) I correct, II incorrect · **Verdict:** Correct

Landfill gas (about 50–60% CH_4 and 40–50% CO_2) comes from the decay of the biodegradable organic fraction, mostly under anaerobic conditions. Glass, plastics and metals are inert.

Q56 · Multiple statements

Choose the correct statements about the composting process.

A. Composting is typically carried out under aerobic conditions.

B. Microorganisms play an important role in composting.

C. Composting is primarily a physical separation process.

D. Organic waste can be converted into compost through biological decomposition.

Options:

1. Only A and B
2. Only B and C
3. Only A, B and D
4. Only A, C and D

Official key: (3) Only A, B and D · **Verdict:** Correct

Composting is aerobic biological decomposition. It is not a physical separation, so C is false.

Q57 · Multiple statements

Consider the following statements about agricultural waste management. Which of the above statements are correct?

Statement A: Crop residues can be used as raw materials for composting, anaerobic digestion and bioenergy production, depending on their characteristics.

Statement B: Open burning of crop residues is an environmentally preferable management method because it eliminates all pollutants.

Statement C: Anaerobic digestion of suitable agricultural organic waste can generate biogas and a digestate requiring appropriate management.

Statement D: Agricultural waste management should consider both resource recovery potential and the environmental impacts of disposal methods.

Options:

1. Statements A, B, C and D
2. Statements A and B only
3. Statements B and C only
4. Statements A, C and D only

Official key: (4) A, C and D only · **Verdict:** Correct

Open burning releases PM2.5, CO and other pollutants, so B is false.

Q58 · Sequence

Which sequence correctly represents the temperature-related progression described for aerobic composting?

Options:

1. Thermophilic microorganisms → temperature rise → mesophilic activity → temperature fall → compost maturation
2. Bacteria and fungi → temperature rise → thermophilic organisms → temperature fall → fungi and heat-sensitive bacteria
3. Temperature rise → mesophilic microorganisms → temperature fall → thermophilic activity → compost maturation
4. Mesophilic microorganisms → temperature fall → thermophilic activity → temperature rise → compost maturation

Official key: (2) · **Verdict:** Correct

Mesophilic bacteria and fungi raise the temperature, and thermophiles take over at about 45–65 °C, which kills pathogens. As the pile cools, mesophiles and fungi recolonise it.

Q59 · Analogy

Which of the following options correctly completes the pair related to the chemical characteristics of solid waste?

Carbon content : CO_2 Emission estimation :: Water content : ?

Options:

1. Influence on calorific value
2. Identification of waste source
3. Determination of collection route
4. Classification by administrative authority

Official key: (1) Influence on calorific value · **Verdict: Correct**

Moisture lowers the net heating value of waste, which is the key property for incineration and RDF.

Q60 · Match the following

Match Column I (Domestic Waste Management Situation) with Column II (Most appropriate classification or management implication) and select the correct combination from the options given below.

(Column I) Statement	(Column II) Implication
A. Vegetable scraps separated before mixing with other refuse	I. Dry recyclable waste, provided it is clean and accepted by the recycling system
B. Glass bottles mixed with food residues	II. Source-separated biodegradable fraction suitable for biological processing
C. Clean plastic packaging discarded by a household	III. Recyclable material whose recovery may be reduced by contamination
D. Discarded household fluorescent lamps	IV. Special waste requiring separate handling because of potentially hazardous components

Options:

1. A-I, B-II, C-III, D-IV
2. A-II, B-I, C-IV, D-III
3. A-II, B-III, C-I, D-IV
4. A-IV, B-III, C-II, D-I

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Official key: (3) A-II, B-III, C-I, D-IV · **Verdict: Correct**

CFLs and tube lights contain mercury, so they count as domestic hazardous waste.

Q61–75 · Process Calculations & Thermodynamics

Q61 · Numerical

A closed system receives 50 kJ of heat and does 20 kJ of work on the surroundings. What is the change in its internal energy?

Options:

1. 70 kJ
2. −30 kJ
3. 30 kJ
4. −70 kJ

Official key: (3) 30 kJ · **Verdict:** Correct

$$\Delta U = Q - W = 50 - 20 = 30\ \text{kJ}$$

Q62 · Standard MCQ

Which of the following statements is NOT consistent with the Second Law of Thermodynamics?

Options:

1. Entropy of an isolated system can remain constant for a reversible process
2. Entropy of an isolated system can increase for an irreversible process
3. Entropy of an isolated system must always decrease
4. Entropy can decrease in one part of a system while increasing elsewhere

Official key: (3) · **Verdict:** Correct

For an isolated system, ΔS ≥ 0, so a forced decrease is impossible.

Q63 · Standard MCQ

A liquid mixture of benzene and toluene is at a fixed pressure. Which statement is correct for the vapour–liquid equilibrium region?

Options:

1. The temperature and phase compositions can vary
2. Only the temperature can vary
3. Only the pressure can vary
4. The phase compositions must remain equal

Official key: (1) · **Verdict:** Correct

By the phase rule, $F = 2 - \pi + N = 2$. With P fixed, one degree of freedom remains, so T and the compositions vary together along the Txy envelope. The phase compositions differ unless the system is at an azeotrope.

Q64 · Standard MCQ

Which of the following is not a fundamental physical quantity in engineering measurements?

Options:

1. Velocity, as it depends on length and time combined
2. Length, as it measures spatial extent in defined units
3. Time, as it measures the duration of physical processes
4. Temperature, as it defines thermal state of a substance

Official key: (1) Velocity · **Verdict:** Correct

Velocity, LT^{-1}, is a derived quantity. Length, time and temperature are SI base quantities.

Q65 · Standard MCQ

Which of the following statements correctly defines a residual thermodynamic property of a real fluid?

Options:

1. Property unchanged when the temperature and pressure of a fluid are varied
2. Property defined only for substances at their critical point conditions
3. Difference between a real fluid and an ideal gas at the same T and P
4. Difference between a liquid and a vapor at equilibrium conditions

Official key: (3) · **Verdict:** Correct

A residual property is $M^R = M - M^{ig}$, both evaluated at the same T and P. Option 4 describes a property change of vaporisation.

Q66 · Standard MCQ

For a nonideal solution, the activity coefficient of component i is greater than 1. What does this indicate about its activity relative to the corresponding ideal-solution value at the same temperature, pressure, and composition?

Options:

1. The activity is equal to the ideal-solution value
2. The activity is less than the ideal-solution value
3. The activity is independent of the composition
4. The activity is greater than the ideal-solution value

Official key: (4) · **Verdict:** Correct

Since $a_i = \gamma_i x_i$, $\gamma_i > 1$ gives $a_i > x_i$. This is a positive deviation from Raoult's law.

Q67 · Numerical

A binary mixture has a total molar volume of 100 cm^3/mol. The mole fractions of components 1 and 2 are 0.4 and 0.6, respectively. The partial molar volume of component 1 is 80 cm^3/mol. What is the partial molar volume of component 2?

Options:

1. 20 cm^3/mol
2. 100 cm^3/mol
3. 113.3 cm^3/mol
4. 133.3 cm^3/mol

Official key: (3) 113.3 cm^3/mol · **Verdict:** Correct

$$V = x_1\bar{V}_1 + x_2\bar{V}_2$$

$$100 = 0.4(80) + 0.6\,\bar{V}_2 \;\Rightarrow\; \bar{V}_2 = \frac{68}{0.6} = 113.3\ \text{cm}^3/\text{mol}$$

Q68 · Multiple statements

Consider the following statements for a binary system at vapor–liquid equilibrium and choose the INCORRECT statement/s.

i. The temperature of the vapor and liquid phases is the same.

ii. The pressure of the vapor and liquid phases is the same.

iii. The chemical potential of each component is the same in both phases.

iv. The mole fraction of each component is the same in both phases.

Options:

1. i and ii only
2. ii and iv only
3. iv only
4. iii and iv only

Official key: (3) iv only · **Verdict:** Correct

At equilibrium, T, P and μ_i (or the fugacity) of each component are equal across phases, but $y_i \neq x_i$ in general.

Q69 · Multiple statements

For predicting vapor–liquid equilibrium of a non-ideal liquid mixture at relatively low pressure, consider the following statements and choose the INCORRECT statement/s.

i. The vapor phase may be treated as an ideal gas.

ii. The liquid-phase nonideality can be represented using activity coefficients.

iii. Raoult's law can be used without modification for a nonideal liquid phase.

iv. The vapor–liquid equilibrium calculation requires consideration of the liquid-phase activity.

Options:

1. i and ii only
2. iii only
3. ii and iii only
4. i and iv only

Official key: (2) iii only · **Verdict:** Correct

The modified Raoult's law, $y_i P = x_i \gamma_i P_i^{sat}$, is needed for a nonideal liquid.

Q70 · Two statements

Read the statements carefully and choose the correct answer.

Statement I: Individual component balances can be formulated separately for each species in a multicomponent flowing system.

Statement II: A total mass balance alone is sufficient to determine the flow rates of individual components in every multicomponent process.

Options:

1. Statement I is incorrect; II is correct
2. Both statements are incorrect
3. Both statements I and II are correct
4. Statement I is correct; II is incorrect

Official key: (4) · **Verdict:** Correct

N components give N independent balances, so the total balance alone cannot fix the individual component flows.

Q71 · Multiple statements

Read the following statements carefully and select the correct option.

Statement I: Reaction progress can be represented using the extent of reaction.

Statement II: The Gibbs free energy of a reacting system can be examined as a function of reaction extent.

Statement III: At constant temperature and pressure, equilibrium corresponds to a nonzero Gibbs-energy derivative with respect to reaction extent.

Options:

1. Statements I and III are true; Statement II is false.
2. Statements I and II are true; Statement III is false.
3. Statements I, II and III are all true.
4. Statements II and III are true; Statement I is false.

Official key: (2) · **Verdict:** Correct

At equilibrium, $(\partial G^t/\partial \varepsilon)$ at constant T and P equals 0, not a nonzero value.

Q72 · Match the following

Match each item in Column A with the correct item in Column B and select the correct option.

Column A: Condition	Column B: Interpretation
A. Steady-state operation	1. Outlet contribution exceeds inlet contribution
B. Unsteady-state operation	2. Accumulation rate is zero
C. Positive accumulation	3. Inventory changes with time
D. Negative accumulation	4. Inlet contribution exceeds outlet contribution

Options:

1. A - 4; B - 1; C - 2; D - 3
2. A - 3; B - 2; C - 1; D - 4
3. A - 2; B - 4; C - 3; D - 1
4. A - 2; B - 3; C - 4; D - 1

Official key: (4) A-2, B-3, C-4, D-1 · **Verdict:** Correct

The general balance is accumulation = in − out + generation − consumption.

Q73 · Multiple statements

Which of the following combinations correctly identifies the behavior of a reacting system at constant temperature and pressure?

1. For dG<0: reaction proceeds toward products
2. For dG=0: equilibrium composition is established
3. For dG=0: reaction proceeds spontaneously forward
4. Extent of reaction represents equilibrium composition directly

Options:

1. Only 1 and 2
2. Only 2 and 4
3. Only 1 and 3
4. Only 1, 2 and 4

Official key: (1) Only 1 and 2 · **Verdict:** Correct

Statement 3 contradicts statement 2. Statement 4 is false because the extent of reaction measures progress, and only its equilibrium value ε_e corresponds to the equilibrium composition.

Q74 · Ranking

Which of the following sequences correctly arranges the states in increasing order of vapour-phase presence?

Options:

1. Saturated vapour < Saturated liquid < Two-phase mixture < Compressed liquid
2. Two-phase mixture < Saturated liquid < Compressed liquid < Saturated vapour
3. Compressed liquid < Saturated liquid < Two-phase mixture < Saturated vapour
4. Saturated liquid < Compressed liquid < Saturated vapour < Two-phase mixture

Official key: (3) · **Verdict:** Correct

Quality x is 0 for any liquid, between 0 and 1 for a wet mixture, and 1 for saturated vapour. A compressed (subcooled) liquid is furthest from vaporisation, so it comes first.

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Q75 · Ranking

Which sequence shows increasing importance of intermolecular effects in real-gas behaviour?

Options:

1. Weak real-gas effects → Ideal-gas behaviour → Pronounced non-ideality → Strong real-gas effects
2. Ideal-gas behaviour → Strong real-gas effects → Weak real-gas effects → Pronounced non-ideality
3. Pronounced non-ideality → Strong real-gas effects → Weak real-gas effects → Ideal-gas behaviour
4. Ideal-gas behaviour → Weak real-gas effects → Strong real-gas effects → Pronounced non-ideality

Official key: (4) · **Verdict:** Correct

As pressure rises or temperature falls, Z departs further from 1 and intermolecular forces dominate.

Q76–91 · Ecosystems

Q76 · Standard MCQ

Which principle is essential for understanding the flow of matter through ecological systems?

Options:

1. Energy transfer connects biotic and abiotic system parts
2. Population growth connects biotic and abiotic system parts
3. Cycling of nutrients connects biotic and abiotic system parts
4. Species migration connects biotic and abiotic system parts

Official key: (3) · **Verdict:** Correct

Matter cycles through the biogeochemical cycles, while energy flows one way. The question asks about matter.

Q77 · Standard MCQ

What is a common impact of human settlements on natural ecosystems?

Options:

1. Maintenance of undisturbed landscapes
2. Restoration of untouched wilderness
3. Enhanced connectivity with natural ecosystem
4. Alteration of natural habitats and resources

Official key: (4) · **Verdict:** Correct

Settlements convert and fragment habitats and draw down resources.

Q78 · Standard MCQ

In a food web, what is the role of decomposers regarding nutrient cycling?

Options:

1. Decomposers convert organic matter into inorganic nutrients
2. Decomposers transfer solar energy toward higher trophic levels
3. Decomposers consume only living organisms for their nutrition
4. Decomposers generate chemical energy for other organisms

Official key: (1) · **Verdict:** Correct

Decomposers carry out mineralisation, releasing inorganic nutrients such as NH_4^+ and PO_4^{3-} for producers to reuse.

Q79 · Standard MCQ

When comparing ecological pyramids, which pyramid best illustrates the principle that energy decreases at higher trophic levels and cannot be inverted?

Options:

1. Pyramid of numbers
2. Pyramid of biomass
3. Pyramid of productivity
4. Pyramid of energy

Official key: (4) Pyramid of energy · **Verdict:** Correct

Only about 10% of energy passes up each level (Lindeman), so the energy pyramid is always upright.

Q80 · Standard MCQ

How does the presence of omnivores in a food web influence ecosystem stability compared with a web containing only strict herbivores and carnivores?

Options:

1. Omnivory always destabilizes ecosystems by increasing trophic competition
2. Omnivory eliminates predator–prey interactions across different trophic levels
3. Omnivory can stabilize ecosystems by providing alternative feeding pathways
4. Omnivory prevents population fluctuations by maintaining constant energy transfer

Official key: (3) · **Verdict:** Correct

Omnivores add links between levels, which gives the web redundancy and buffers it against the loss of one prey.

Q81 · Assertion–Reason

Directions: Each question consists of an assertion followed by a reason. Select the correct answer using the codes below.

Assertion (A): Bioaccumulation can affect the reproductive success of organisms at higher trophic levels.

Reason (R): Fat-soluble toxins stored in tissues may disrupt endocrine function over time.

Options:

1. Both A and R are correct, and R explains A.
2. Both A and R are correct, but R does not explain A.
3. A is correct, but R is incorrect.
4. A is incorrect, but R is correct.

Official key: (1) · **Verdict:** Correct

DDT disrupts calcium metabolism and hormones, causing eggshell thinning and reproductive failure in birds of prey.

Q82 · Assertion–Reason

Direction: Each question consists of two statements, Assertion (A) and Reason (R). Choose the correct option:

A: Biomagnification can result in harmful effects to apex predators in an ecosystem.

R: Apex predators occupy positions in the food chain where toxin concentrations become diluted due to metabolic breakdown.

Options:

1. Both A and R are correct, and R explains A.
2. Both A and R are correct, but R does not explain A.
3. A is correct, but R is incorrect.
4. A is incorrect, but R is correct.

Official key: (3) A correct, R incorrect · **Verdict:** Correct

Concentrations increase, not dilute, at higher trophic levels. In NCERT's DDT example they rise from 0.003 ppb in water to 25 ppm in fish-eating birds.

Q83 · Assertion–Reason

Direction: Each question consists of an Assertion (A) and a Reason (R). Choose the correct option:

Assertion: In a terrestrial food web, the loss of a primary consumer species can destabilize energy flow across multiple trophic levels.

Reason: Primary consumers directly transfer energy from producers to higher trophic organisms, and their removal alters ecosystem structure.

Options:

1. Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).
2. Both Assertion (A) and Reason (R) are true, but Reason (R) is NOT the correct explanation of Assertion (A).
3. Assertion (A) is true but Reason (R) is false.
4. Assertion (A) is false but Reason (R) is true.

Official key: (1) · **Verdict:** Correct

Herbivores are the only bridge between producers and higher consumers, so losing them disrupts every level above.

Q84 · Two statements

Consider the following statements about energy flow in ecosystems:

Statement I: Energy flow through an ecosystem is linear.

Statement II: Energy is progressively converted into heat as it passes through tropic levels.

Which of the following is correct?

Options:

1. Statement I only
2. Statement II only
3. Both Statement I and Statement II
4. Neither Statement I nor Statement II

Official key: (3) Both · **Verdict:** Correct

Energy flow is unidirectional, and it is lost as heat at every step.

Q85 · Statement–explanation

Which option correctly evaluates the statements about ecosystem components?

Statement I: Dead leaf litter is an abiotic component even though it originated from plants.

Statement II: Decomposer microorganisms convert organic matter into simpler substances available to producers.

Options:

1. Both Statements I and II are correct, and Statement II explains Statement I
2. Both Statements I and II are correct, but Statement II does not explain Statement I
3. Statement I is correct, but Statement II is incorrect
4. Statement I is incorrect, but Statement II is correct

Official key: (4) Statement I incorrect, II correct · **Verdict:** Very likely wrong. The better answer is (2). Challenge this key.

Statement II is plainly true. The dispute is Statement I. Standard ecology texts divide an ecosystem into living (biotic) and non-living (abiotic) parts, and place dead organic matter (detritus, litter, humus) on the non-living side. Odum lists the abiotic components as inorganic substances, **organic compounds (proteins, carbohydrates, lipids, humic substances) that link biotic and abiotic**, and the climate regime. Dead leaf litter is no longer living, so Statement I is true.

Statement II describes what decomposers do. It does not explain why litter is abiotic, so (2) follows. The examiners likely meant "litter is still biotic because it came from a living source". That view is not supported by the standard texts. Full references are in "Wrong and disputed keys" at the end.

Reference: E.P. Odum & G.W. Barrett, *Fundamentals of Ecology*, chapter on the ecosystem (Components of an ecosystem); Erach Bharucha, *Textbook of Environmental Studies*, chapter on ecosystems (Structure: abiotic components include organic matter); NCERT Biology XII, chapter "Ecosystem" (Detritus as dead organic matter).

Q86 · Multiple statements

Which of the following statements correctly describe trophic energy flow?

I. Producers convert radiant energy into chemical energy.

II. Food webs contain interconnected food chains.

III. Energy is lost as decomposers recycle heat.

Options:

1. I only
2. II and III only
3. I and II only
4. I, II and III

Official key: (4) I, II and III · **Verdict:** Wrong. The correct answer is (3), I and II only. Challenge this key.

I and II are textbook statements. III is scientifically false: heat cannot be recycled by decomposers or by any organism. Decomposers recycle nutrients. Their respiration releases energy as heat, which leaves the ecosystem for good, so energy flow is one-way. Accepting III would mean energy cycles, which contradicts both the second law and the basic rule that matter cycles and energy flows. The key is wrong, and full references are in "Wrong and disputed keys" at the end.

Reference: NCERT Biology XII, chapter "Ecosystem" (Energy flow: unidirectional; Nutrient cycling); Odum, chapter on energy in ecological systems (laws of thermodynamics).

Q87 · Multiple statements

Which of the following statements correctly describe ecological succession?

I. Primary succession begins where developed soil is initially absent.

II. Secondary succession generally proceeds faster because soil or propagules remain.

III. Succession always proceeds without disturbance or retrogression.

Options:

1. I only
2. I and II only
3. II and III only
4. I, II and III

Official key: (2) I and II only · **Verdict:** Correct

Disturbances such as fire or deforestation can reset succession, so III is false.

Q88 · Match the following

Which option correctly matches each ecosystem term with its role?

List I	List II
P. Standing crop	1. Rate of biomass formation
Q. Standing state	2. Organisms present per unit area
R. Productivity	3. Nutrient quantity at a given time
S. Decomposition	4. Breakdown of detritus

Options:

1. P–2, Q–3, R–1, S–4
2. P–3, Q–2, R–4, S–1
3. P–1, Q–4, R–2, S–3
4. P–4, Q–1, R–3, S–2

Official key: (1) · **Verdict:** Correct

Standing crop is the living material per unit area, and standing state is the nutrients present at a given time.

Q89 · Correct pair

Which contaminant–property pair supports biomagnification?

A. Nitrate — high lipid storage

B. Methylmercury — strong tissue retention

C. Glucose — environmental persistence

D. Ammonium — trophic stability

Options:

1. A
2. B
3. C
4. D

Official key: (2) B · **Verdict:** Correct

Methylmercury is persistent, binds to tissue and is excreted slowly, as the Minamata episode showed.

Q90 · Sequence

Which sequence represents primary succession on exposed rock?

A. Shrubs → herbs → moss → lichen

B. Moss → lichen → shrubs → herbs

C. Herbs → shrubs → lichen → moss

D. Lichen → moss → herbs → shrubs

Options:

1. A
2. B
3. C
4. D

Official key: (4) D · **Verdict:** Correct

On bare rock (a xerosere), lichens come first, then mosses, then herbs, then shrubs, and finally forest.

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Q91 · Sequence

Which sequence correctly traces solar energy through an ecosystem?

1. Herbivore ingestion
2. Producer assimilation
3. Carnivore ingestion
4. Respiratory dissipation

A. 2 → 1 → 3 → 4

B. 1 → 2 → 4 → 3

C. 2 → 3 → 1 → 4

D. 4 → 2 → 1 → 3

Options:

1. A
2. B
3. C
4. D

Official key: (1) A · **Verdict:** **Correct**

Energy passes from producers to herbivores to carnivores and is finally lost as heat.

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Q92–106 · Natural Resources

This block follows NCERT Class X Geography, *Contemporary India II*, closely, often word for word. It draws on the chapters "Resources and Development", "Water Resources" and "Minerals and Energy Resources".

Q92 · Standard MCQ

Which basis distinguishes biotic resources from abiotic resources?

Options:

1. Basis of resource exhaustibility
2. Basis of resource development
3. Basis of resource origin
4. Basis of resource ownership

Official key: (3) Origin · **Verdict:** Correct

NCERT classifies resources by origin (biotic or abiotic), exhaustibility (renewable or non-renewable), ownership (individual, community, national, international) and status of development (potential, developed, stock, reserves).

Q93 · Standard MCQ

Which agricultural activity places substantial pressure on water resources?

Options:

1. Expansion of irrigated areas
2. Cultivation of drought-resistant crops
3. Reduction of agricultural irrigation
4. Development of dry farming

Official key: (1) · **Verdict:** Correct

Irrigation is the largest single use of water.

Q94 · Standard MCQ

Under the resource classification framework, what is the primary criterion used to distinguish between renewable and non-renewable resources?

Options:

1. According to their resource origin
2. According to their resource ownership
3. According to their resource exhaustibility
4. According to their development status

Official key: (3) Exhaustibility · **Verdict:** Correct

Renewable resources, such as solar energy, water, forests and wildlife, are replenished by natural processes. Non-renewable resources, such as minerals and fossil fuels, take millions of years to form and are used up.

Q95 · Standard MCQ

What is the impact of intensive irrigation practices on groundwater availability and food security?

Options:

1. Raises groundwater and improves food security
2. Lowers groundwater and threatens food security
3. Maintains groundwater and supports food security
4. Reduces water use and improves food security

Official key: (2) · **Verdict:** Correct

Over-pumping lowers water tables and puts food security at risk.

Q96 · Standard MCQ

Which classification best describes forests and wildlife based on their biological origin and ability to regenerate naturally?

Options:

1. Renewable resources supported by natural biological processes
2. Non-renewable resources formed through geological processes
3. Mineral resources obtained from geological formations
4. Abiotic resources derived from physical and chemical processes

Official key: (1) · **Verdict:** Correct

Flora and fauna are NCERT's own examples of biological renewable resources.

Q97 · Standard MCQ

How can excessive dependence on private wells affect long-term agricultural water security?

Options:

1. Lowers groundwater and threatens supply
2. Raises groundwater and improves supply
3. Reduces pumping and lowers groundwater
4. Raises groundwater but reduces supply

Official key: (1) · **Verdict:** Correct

Many private wells pump without any regulation, so aquifers are drawn down faster than they recharge. Water tables fall and long-term farm supply becomes insecure.

Q98 · Standard MCQ

How does expanding irrigation during dry seasons affect water resources?

Options:

1. Increases irrigation water extraction
2. Reduces demand through drought-resistant crops
3. Prevents extraction through dry farming
4. Maintains groundwater through lower demand

Official key: (1) · **Verdict:** Correct

Dry-season rainfall cannot meet crop needs, so expanding irrigation then must draw more water from reservoirs and groundwater.

Q99 · Multiple statements

Consider the following statements regarding mineral resources:

Statement I: Metallic minerals generally contain economically recoverable metallic elements within their composition.

Statement II: Non-metallic minerals generally lack economically recoverable metallic elements within their composition.

Statement III: Metallic minerals are commonly classified into ferrous and non-ferrous mineral groups.

Which of the statements given above is/are correct?

Options:

1. Statement I only
2. Statements I and II only
3. Statements II and III only
4. Statements I, II and III

Official key: (4) All three · **Verdict:** Correct

Ferrous minerals include iron ore and manganese. Non-ferrous minerals include copper and bauxite.

Q100 · Two statements

Read the statements carefully and choose the correct answer.

Statement I: Excessive use of wells and tube-wells for irrigation can reduce groundwater levels.

Statement II: Falling groundwater levels can affect food security.

Options:

1. Statement I is correct; II is incorrect
2. Both statements are incorrect
3. Statement I and II are correct
4. Statement I is incorrect; II is correct

Official key: (3) Both correct · **Verdict:** Correct

NCERT notes that over-exploitation of groundwater through wells and tube-wells has lowered water tables, which in turn affects water availability and food security.

Q101 · Two statements

Read the statements carefully and choose the correct answer.

Statement I: The first spell of rain was collected in traditional rooftop harvesting systems.

Statement II: Rainfall water was collected after the roofs and pipes had been cleaned.

Options:

1. Statement I is incorrect; II is correct
2. Both statements are incorrect
3. Statement I is correct; II is incorrect
4. Statement I and II are correct

Official key: (1) I incorrect, II correct · **Verdict:** Correct

NCERT, describing the tankas of Bikaner and Phalodi, says the first spell of rain was usually not collected because it cleaned the roofs and pipes. Water from later showers was stored.

Q102 · Multiple statements

Read the following statements carefully and select the correct option.

Statement I: Urbanisation increases demands for both water and energy resources.

Statement II: Urban groundwater pumping contributes to freshwater resource depletion.

Statement III: Dense urban populations reduce pressure on existing freshwater resources.

Options:

1. Statements I and II are true; Statement III is false.
2. Statements I, II and III are all true.
3. Statements I and III are true; Statement II is false.
4. Statements II and III are true; Statement I is false.

Official key: (1) · **Verdict:** Correct

Cities raise demand for both water and energy, and urban groundwater pumping depletes aquifers. Dense populations increase pressure on freshwater, so Statement III is false.

Q103 · Match the following

Match each item in Column A with the correct item in Column B and select the correct option.

Column A: Water-related condition	Column B: Associated explanation
A. Large population	1. Causes declining groundwater levels
B. Water pollution	2. Makes available water hazardous
C. Irrigated agriculture	3. Remains the largest water consumer
D. Groundwater overuse	4. Creates greater demand for water

Options:

1. A - 4; B - 2; C - 3; D - 1
2. A - 2; B - 1; C - 4; D - 3
3. A - 3; B - 4; C - 1; D - 2
4. A - 4; B - 3; C - 2; D - 1

Official key: (1) · **Verdict:** Correct

A large population drives water demand, pollution makes available water hazardous, irrigated agriculture is the largest consumer, and groundwater overuse lowers water tables.

Q104 · Multiple statements

Which pairs of combination correctly identify the purposes associated with modern dams?

1. Irrigation: Impounded water supports agricultural requirements
2. Hydroelectricity: Impounded water supports electricity generation
3. Flood control: Regulated water flow helps control floods
4. Industrial supply: Dams exclude water provision for industries

Options:

1. Only 2, 3 and 4
2. Only 1, 2 and 4
3. Only 1, 2 and 3
4. Only 1, 3 and 4

Official key: (3) Only 1, 2 and 3 · **Verdict:** Correct

NCERT lists irrigation, electricity, domestic and industrial water supply, flood control, recreation, inland navigation and fish breeding as uses of dams. Item 4 is therefore false.

Q105 · Sequence

Which sequence correctly represents the order of water availability stages for traditional rooftop harvesting as described in the context?

Options:

1. First rainfall → Roof cleaning → Subsequent rainfall → Tanka storage
2. Roof cleaning → First rainfall → Tanka storage → Subsequent rainfall
3. Subsequent rainfall → First rainfall → Roof cleaning → Tanka storage
4. Tanka storage → Subsequent rainfall → First rainfall → Roof cleaning

Official key: (1) · **Verdict:** Correct

The stem refers to “the context”, a passage the paper does not print. This is the NCERT tanka description, consistent with Q101.

Q106 · Sequence

Arrange the following processes involved in residual ore formation in their correct chronological sequence:

A. Leaching removes soluble constituents from weathered rocks

B. Chemical weathering decomposes minerals in parent rocks

C. Valuable minerals become concentrated within residual materials

D. Residual materials accumulate after extensive leaching

Options:

1. A → B → C → D
2. B → A → D → C
3. A → C → B → D
4. C → B → A → D

Official key: (2) B → A → D → C · **Verdict:** Correct

Rocks weather, soluble parts leach away, the residue accumulates and the ore becomes concentrated, as with bauxite.

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Q107–121 · Biodiversity & Conservation

Q107 · Standard MCQ

Which statement describes biodiversity as defined in environmental science?

Options:

1. The total population size of organisms within an ecosystem
2. The variety of life occurring on Earth
3. The quantity of biological resources available for human use
4. The distribution of organisms across a particular geographical region

Official key: (2) · **Verdict:** Correct

Biodiversity is the variability among living organisms, covering genetic, species and ecosystem diversity.

Q108 · Standard MCQ

Which of the following combinations consists entirely of ecosystem types used to illustrate ecosystem diversity?

Options:

1. Populations, individuals and genes
2. Genes, alleles and chromosomes
3. Birds, mammals and reptiles
4. Forests, grasslands and deserts

Official key: (4) · **Verdict:** Correct

Ecosystem diversity is the variety of habitats, such as forests, grasslands, deserts and wetlands. Genes and alleles are genetic diversity, and animal groups are species diversity.

Q109 · Standard MCQ

Which of the following values supports conserving biodiversity to ensure resources for future generations?

Options:

1. Ethical value
2. Aesthetic value
3. Option value
4. Consumptive value

Official key: (3) Option value · **Verdict:** Correct

Option value means keeping biodiversity for unknown future uses, such as new drugs and crop genes.

Q110 · Standard MCQ

Two breeding populations of the same widely distributed species exhibit distinct patterns of genetic polymorphism. What does this suggest?

Options:

1. Reduction of species-level biodiversity
2. Absence of genetic differentiation
3. Geographic variation in genetic diversity
4. Complete genetic uniformity within species

Official key: (3) · **Verdict:** Correct

NCERT's example is *Rauwolfia vomitoria*, whose reserpine content varies across Himalayan ranges.

Q111 · Standard MCQ

Kaziranga National Park is globally recognized for conserving which flagship species through habitat protection?

Options:

1. Indian rhinoceros
2. Snow leopard
3. Bengal tiger
4. Asiatic lion

Official key: (1) · **Verdict:** Correct

Kaziranga holds about two-thirds of the world's greater one-horned rhinos and is a UNESCO World Heritage Site.

Q112 · Standard MCQ

Which combination of factors poses the greatest cumulative risk to a small endemic species?

Options:

1. Large range, high dispersal, and habitat connectivity
2. Stable habitat, high genetic variability, and low disturbance
3. Habitat fragmentation, restricted range, and climate change
4. High reproductive rate, broad tolerance, and low exploitation

Official key: (3) · **Verdict:** Correct

A small endemic species with a restricted range cannot move away from fragmented habitat or a shifting climate, so these threats compound. The other options list traits that protect a species.

Q113 · Standard MCQ

Which ex situ method is most appropriate for conserving genetic diversity of a plant species whose seeds rapidly lose viability during conventional storage?

Options:

1. Long-term seed drying
2. Maintenance in sacred groves
3. Protection within sanctuaries
4. Cryopreservation of tissues

Official key: (4) · **Verdict:** Correct

Recalcitrant seeds cannot be dried, so embryos or tissues are stored at −196 °C. Sacred groves and sanctuaries are in situ methods.

Q114 · Standard MCQ

A recalcitrant-seeded tropical tree species cannot be conserved by seed banking. Which in-vitro method provides secure long-term genetic stability with minimal maintenance?

Options:

1. Field gene bank maintenance
2. Slow-growth storage with low temperature
3. Cryopreservation using liquid nitrogen
4. Frequent subculturing in vitro

Official key: (3) · **Verdict:** Correct

Cryopreservation halts metabolism. Slow growth and subculturing need repeated handling and risk somaclonal variation.

Q115 · Two statements

Consider the following statements about biodiversity:

Statement P: The number of different species in an area can be used to measure biodiversity.

Statement Q: The relative abundance of different species can also be used to assess biodiversity.

Choose the CORRECT answer.

Options:

1. P is TRUE; Q is FALSE.
2. P is FALSE; Q is TRUE
3. P is TRUE; Q is TRUE
4. P is FALSE; Q is FALSE

Official key: (3) · **Verdict:** Correct

Species richness and evenness are the two components of diversity indices such as Shannon's.

Q116 · Two statements

Consider the following statements regarding biodiversity:

Statement P. Biodiversity provides ecosystem services that support human well-being.

Statement Q. Biodiversity contributes to ecosystem stability and resilience.

Choose the CORRECT answer.

Options:

1. Statement P is TRUE; Statement Q is FALSE
2. Statement P is FALSE; Statement Q is TRUE
3. Statement P is TRUE; Statement Q is TRUE
4. Statement P is FALSE; Statement Q is FALSE

Official key: (3) · **Verdict:** Correct

Biodiversity supports provisioning, regulating, cultural and supporting services. Tilman's long-term plots showed that species-rich communities are more stable, so both statements are true.

Q117 · Multiple statements

Which of the following statements about an ecosystem are CORRECT?

A- It includes organisms and their physical environment.

B- Organisms interact with each other and their environment.

C- It is equivalent to a population in an area.

D- It functions independently of its environment.

Options:

1. A, C and D only
2. A and B only
3. B, C and D only
4. B and C only

Official key: (2) A and B only · **Verdict:** Correct

An ecosystem is a community of organisms together with its physical environment, interacting as one unit. It is not a single population (C is false), and it cannot function apart from its environment (D is false).

Q118 · Multiple statements

Consider the following consequences of wetland degradation caused by sewage discharge, nutrient enrichment, and altered water flow:

A. Decline in native aquatic biodiversity

B. Disruption of nutrient cycling

C. Increase in ecological resilience

D. Loss of ecosystem services

Which of the above are CORRECT?

Options:

1. A, C and D only
2. A, B and D only
3. B, C and D only
4. A, B and C only

Official key: (2) A, B and D only · **Verdict:** Correct

Degradation lowers resilience, so C is false.

Q119 · Match the following

Match the following columns:

Animal	Wildlife Sanctuary
A. One-horned rhinoceros	i. Gahirmatha Marine Sanctuary
B. Asiatic lion	ii. Pobitora Wildlife Sanctuary
C. Olive ridley turtle	iii. Gir Wildlife Sanctuary
D. Saltwater crocodile	iv. Bhitarkanika Wildlife Sanctuary

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Options:

1. A–iii, B–ii, C–iv, D–i
2. A–ii, B–iii, C–i, D–iv
3. A–i, B–iii, C–ii, D–iv
4. A–iv, B–i, C–iii, D–ii

Official key: (2) · **Verdict:** Correct

- Pobitora (Assam): the highest density of one-horned rhinos.
- Gir (Gujarat): the Asiatic lion.
- Gahirmatha (Odisha): mass nesting (arribada) of olive ridley turtles.
- Bhitarkanika (Odisha): saltwater crocodiles.

Q120 · Correct pair

Which of the following terms is correctly matched with its description?

Options:

1. Biopiracy — Unauthorised bioresources exploitation
2. Bioremediation — Species trade regulation
3. Bioprospecting — Ozone layer protection
4. Biomagnification — Soil fertility improvement

Official key: (1) · **Verdict:** Correct

Biopiracy is the use of bioresources or traditional knowledge without consent or benefit-sharing, as in the basmati and neem cases.

Q121 · Ranking

Arrange the following in increasing order of spatial scale of species diversity:

A. Gamma diversity

B. Alpha diversity

C. Beta diversity

Options:

1. B < C < A
2. A < B < C
3. C < A < B
4. B < A < C

Official key: (1) Alpha < Beta < Gamma · **Verdict:** Correct

Alpha diversity is within a community, beta is between communities, and gamma is across a region (Whittaker).

Q122–136 · Global Issues & Environmental Law

Q122 · Standard MCQ

Which of the following land-use practices is a major human cause of desertification in India?

Options:

1. Afforestation
2. Soil conservation
3. Crop rotation
4. Overgrazing

Official key: (4) Overgrazing · **Verdict: Correct**

Overgrazing strips the vegetation cover and exposes soil to wind and water erosion. The other options are remedies.

Q123 · Standard MCQ

Which of the following term is explicitly defined under the Environment (Protection) Act, 1986?

Options:

1. Pollutant
2. Biodiversity
3. Ecosystem service
4. Ecological footprint

Official key: (1) Pollutant · **Verdict: Correct**

Section 2(b) defines “environmental pollutant”. None of the other three terms appears in the Act.

Q124 · Standard MCQ

Which of the following waste streams should a waste generator separately store under the Solid Waste Management Rules, 2016?

Options:

1. Mixed waste
2. Biodegradable waste
3. Street sweepings
4. Untreated sludge

Official key: (2) Biodegradable waste · **Verdict: Correct**

Generators must store three streams separately: biodegradable, non-biodegradable and domestic hazardous.

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Q125 · Standard MCQ

According to the Bio-Medical Waste Management Rules 2016, which segregation combination should a vaccination centre use for detached needles and discarded syringes?

Options:

1. Blue needles, Yellow syringes
2. Yellow needles, Blue syringes
3. White needles, Red syringes
4. Red needles, White syringes

Official key: (3) White needles, Red syringes · **Verdict:** Correct

Sharps, including needles, go in the white (translucent) puncture-proof container. Contaminated recyclables, including syringes without needles, go in red bags.

Q126 · Standard MCQ

Which section under the Wild Life (Protection) Act, 1972, must be referred to first when assessing if hunting of a protected wild animal is legally prohibited?

Options:

1. Section 8
2. Section 12
3. Section 9
4. Section 11

Official key: (3) Section 9 · **Verdict:** Correct

Section 9 prohibits hunting. Sections 11 and 12 are the exceptions, covering dangerous animals and special-purpose permits.

Q127 · Standard MCQ

A dryland region develops moving sand dunes and sparse vegetation. Which management approach most effectively halts this desertification process?

Options:

1. Excess irrigation
2. Shelterbelts
3. Dune stabilisation
4. Crop rotation

Official key: (3) Dune stabilisation · **Verdict:** Correct, though (2) can be argued (see "Wrong and disputed keys")

Moving dunes are stopped directly by fixing the sand: micro-windbreaks or checkerboards, mulching, and sand-binding grasses and shrubs. Shelterbelts protect fields from wind behind them but do not fix an active dune. "Dune stabilisation" is the most specific answer, so an objection here is unlikely to succeed.

Reference: Bharucha, chapter on natural resources (Desertification); CAZRI, Jodhpur, technical bulletins on sand-dune stabilisation.

Q128 · Standard MCQ

According to the Plastic Waste Management Rules 2016, which authority is primarily responsible for implementing provisions related to plastic waste in rural areas governed by a Gram Panchayat?

Options:

1. Central Board
2. Municipal Authority
3. Urban Secretary
4. Gram Panchayat

Official key: (4) Gram Panchayat · **Verdict:** Correct

Under Rule 7, each Gram Panchayat sets up, operates and coordinates the plastic waste management system in its rural area, either directly or through an agency.

Q129 · Two statements

Read the statements carefully and choose the correct answer.

Statement I: Increased fossil-fuel use raises atmospheric carbon dioxide levels and disturbs balance.

Statement II: Greenhouse gases release outgoing heat while blocking incoming solar radiation.

Options:

1. Statement I is correct; II is incorrect
2. Statement I is incorrect; II is correct
3. Both statements are incorrect
4. Statement I and II are correct

Official key: (1) · **Verdict:** Correct

Greenhouse gases let incoming shortwave radiation through and trap outgoing longwave radiation. Statement II reverses this.

Q130 · Two statements

Read the statements related to Solid Waste Management Rules, 2016, carefully and choose the correct answer.

Statement I: Waste generators must segregate waste into three separate streams at source.

Statement II: Waste generators must hand over segregated waste to authorised waste pickers or collectors.

Options:

1. Statement I is correct; Statement II is incorrect
2. Statement I and II are correct
3. Statement I is incorrect; Statement II is correct
4. Both statements are incorrect

Official key: (2) Both correct · **Verdict:** Correct

Rule 4(1)(a) requires segregation into biodegradable, non-biodegradable and domestic hazardous waste. Rule 4(1)(b) requires handing it over to authorised waste pickers or collectors, or to the local body.

Q131 · Multiple statements

Consider the following statements regarding stratospheric ozone:

Statement I: Stratospheric ozone protects Earth from harmful ultraviolet radiation.

Statement II: Ultraviolet radiation splits molecular oxygen into free oxygen atoms.

Statement III: Free oxygen atoms combine with molecular oxygen to produce ozone.

Which of the statements given above are correct?

Options:

1. Statements I and III are correct; Statement II is incorrect.
2. Statements I and II are correct; Statement III is incorrect.
3. Statements I, II and III are all correct.
4. Statements II and III are correct; Statement I is incorrect.

Official key: (3) All correct · **Verdict:** **Correct**

This is the Chapman cycle: O_2 + UV gives 2 O, and O + O_2 gives O_3.

Q132 · Multiple statements

Consider the following statements regarding the Forest (Conservation) Act, 1980.

Statement I: The Act is included among enactments under NGT jurisdiction.

Statement II: The Act concerns substantial questions relating to the environment.

Statement III: The Act is excluded from the enactments listed under NGT jurisdiction.

Which of the statements given above are true?

Options:

1. Statements I and III are true; Statement II is false.
2. Statements I and II are true; Statement III is false.
3. Statements I, II and III are all true.
4. Statements II and III are true; Statement I is false.

Official key: (2) · **Verdict:** **Correct**

The Forest (Conservation) Act is one of the seven Acts in Schedule I of the NGT Act. Statement III contradicts statement I.

Q133 · Match the following

Match each item in Column A with the correct item in Column B and select the correct option.

Column A: Provision	Column B: Corresponding requirement
A. Plastic waste	1. Scientific reduction and recycling
B. Waste management	2. Discarded plastic after intended use
C. Municipal authority	3. Collection and disposal of plastic waste
D. State Pollution Control Board	4. Registration of plastic manufacturers

Options:

1. A - 2; B - 1; C - 3; D - 4
2. A - 4; B - 3; C - 1; D - 2
3. A - 3; B - 4; C - 2; D - 1
4. A - 1; B - 2; C - 4; D - 3

Official key: (1) · **Verdict:** Correct

Rule 3 defines plastic waste and waste management, Rule 6 sets out the municipal authority's duties, and Rule 13 covers registration with the SPCB.

Q134 · Multiple statements

Which pair of statements correctly describe the Precautionary Principle as applied in environmental judicial decisions?

1. Environmental degradation — Preventive measures
2. Serious irreversible damage — Scientific certainty not essential
3. Environmental authorities — Anticipate causes of degradation
4. Scientific uncertainty — Justification for delaying prevention

Options:

1. Only 1, 3 and 4
2. Only 1, 2 and 3
3. Only 2, 3 and 4
4. Only 1, 2 and 4

Official key: (2) Only 1, 2 and 3 · **Verdict:** Correct

In *Vellore Citizens' Welfare Forum*, the Supreme Court held that the state must anticipate, prevent and attack the causes of degradation. Lack of scientific certainty is not a reason to postpone measures, and the onus of proof is on the developer. Item 4 inverts the principle.

Q135 · Sequence

Arrange the following stages in the order of their occurrence in the greenhouse-effect process.

Options:

1. Earth's surface absorbs solar energy → Heat radiates upward → Surface temperature increases → Greenhouse gases trap heat
2. Surface temperature increases → Earth's surface absorbs solar energy → Heat radiates upward → Greenhouse gases trap heat
3. Earth's surface absorbs solar energy → Surface temperature increases → Greenhouse gases trap heat → Heat radiates upward
4. Earth's surface absorbs solar energy → Surface temperature increases → Heat radiates upward → Greenhouse gases trap heat

Official key: (4) · **Verdict:** Correct

The surface absorbs shortwave sunlight and warms up. It then radiates longwave infrared upward, and greenhouse gases absorb and re-radiate part of that heat.

Q136 · Sequence

Arrange the following processes in chronological order and choose the correct option.

A. Earth's surface absorbs solar energy

B. Earth's surface temperature increases

C. Surface heat radiates toward atmosphere

D. Greenhouse gases trap part of heat

Options:

1. B, A, C, D
2. A, C, B, D
3. C, A, B, D
4. A, B, C, D

Official key: (4) A, B, C, D · **Verdict:** Correct

This is the same content as Q135, repeated in the paper.

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Q137–150 · EIA, Environmental Management & Sustainability

Q137 · Standard MCQ

What is a key objective of Environmental Impact Assessment (EIA)?

Options:

1. Renewable promotion
2. Impact prediction
3. Biodiversity monitoring
4. Pollution enforcement

Official key: (2) Impact prediction · **Verdict:** Correct

EIA identifies and predicts the effects of a project before it is approved.

Q138 · Standard MCQ

What is a key objective of an Environmental Management Plan (EMP)?

Options:

1. Impact mitigation
2. Alternative identification
3. Community funding
4. Marketing development

Official key: (1) Impact mitigation · **Verdict:** Correct

The EMP sets out how impacts will be mitigated and monitored, by whom, and at what cost. Identifying alternatives belongs to the EIA itself.

Q139 · Standard MCQ

Why are reference sites important in aquatic environmental impact assessment (EIA)?

Options:

1. Control species introduction
2. Impact baseline comparison
3. Weather pattern prediction
4. Historical climate assessment

Official key: (2) · **Verdict:** Correct

Undisturbed reference (control) sites separate project impacts from natural variation, as in the BACI design.

Q140 · Standard MCQ

Which Environmental Impact Assessment (EIA) outcome contributes to sustainable development?

Options:

1. Resource elimination
2. Development delays
3. Economic prioritization
4. Impact mitigation

Official key: (4) · **Verdict:** Correct

By predicting impacts and building mitigation into the project, EIA lets development go ahead while protecting the environment.

Q141 · Standard MCQ

How does the adoption of sustainable agricultural practices directly contribute to the overall goal of sustainable development?

Options:

1. Ecosystem conservation
2. Monoculture expansion
3. Crop rotation elimination
4. Chemical intensification

Official key: (1) · **Verdict:** Correct

Crop rotation, integrated pest management, organic inputs and efficient water use conserve soil, water and biodiversity for future generations. The other options harm ecosystems.

Q142 · Standard MCQ

A local government wants to promote a low-carbon lifestyle in a region dependent on coal-fired electricity. Which strategy addresses both energy transition and community engagement?

Options:

1. Electricity tariff escalation
2. Residential audit mandates
3. Community solar development
4. Natural gas conversion

Official key: (3) · **Verdict:** Correct

Community solar replaces coal power and directly involves residents as owners or subscribers.

Q143 · Standard MCQ

A government is considering a mining project in a region with high biodiversity and indigenous communities. Applying the precautionary principle, which action should it prioritize?

Options:

1. Consult communities only after project commencement
2. Approve the project immediately to boost economic growth
3. Suspend approval until environmental and social assessments are completed
4. Proceed with limited assessment to save time and resources

Official key: (3) · **Verdict:** Correct

Where serious or irreversible harm is possible, the precautionary principle requires full environmental and social assessment, with community consultation, before approval is given.

Q144 · Two statements

Read the following two statements regarding environmental management tools and select the correct option:

Statement I: An environmental audit systematically evaluates an organization's environmental performance against applicable regulatory and management requirements.

Statement II: An environmental audit primarily involves continuous measurement of environmental parameters such as air quality to identify changes in ambient conditions.

Options:

1. Both statements correct
2. Both statements incorrect
3. Statement I correct
4. Statement II correct

Official key: (3) Statement I correct · **Verdict:** Correct

Continuous ambient measurement is environmental monitoring, not an audit.

Q145 · Statement–explanation

Read the following two statements and select their relationship.

Statement I: Environmental constraints can influence project location, scale, technology, and design, not just mitigation.

Statement II: Environmental planning assesses sensitivity, resource limits, carrying capacity, and impacts before project decisions are made.

Options:

1. Both statements are correct, and II correctly explains I
2. Both statements are correct, but II does not explain I
3. Statement I is correct, but Statement II is incorrect
4. Statement I is incorrect, but Statement II is correct

Official key: (1) · **Verdict:** Correct

Because environmental planning comes before project decisions, environmental constraints can shape siting, scale and design.

Q146 · Multiple statements

Which statements regarding the Terms of Reference (ToR) in Environmental Impact Assessment (EIA) are correct?

1. Defines study scope.
2. Identifies key issues.
3. Guides report preparation.
4. Post-operational preparation only.

Options:

1. 1 and 2 only
2. 1, 2 and 3 only
3. 2, 3 and 4 only
4. 1, 2, 3 and 4

Official key: (2) 1, 2 and 3 only · **Verdict: Correct**

The ToR is fixed at the scoping stage, before the study starts.

Q147 · Match the following

Match the column I with column II

Column I	Column II
1. Advertising-driven demand	P. Resource-intensive convenience systems
2. Disposable consumption	Q. Status signaling via material goods
3. Convenience-driven consumption	R. Non-functional preference creation
4. Social signalling	S. Reduced product life extension

Options:

1. 1–R, 2–S, 3–P, 4–Q
2. 1–Q, 2–P, 3–S, 4–R
3. 1–S, 2–R, 3–Q, 4–P
4. 1–P, 2–Q, 3–R, 4–S

Official key: (1) · **Verdict: Correct**

Advertising creates non-functional wants, disposable goods shorten product life, convenience depends on resource-intensive systems, and social signalling uses goods to show status.

Q148 · Multiple statements

Which pair correctly matches the type of objective with its corresponding Environmental Impact Assessment (EIA) goal?

Type of objective	Goal
1. Immediate	Improve the environmental design of the proposal and identify mitigation measures
2. Ultimate/Long-term	Promote sustainable development by preventing irreversible damage to critical resources and livelihoods
3. Procedural	Establish baseline conditions solely for post-project auditing
4. Participatory	Restrict public input to the final public hearing stage only

Options:

1. 1 and 2 only
2. 1, 2 and 3 only
3. 2 and 4 only
4. 1, 2, 3 and 4

Official key: (1) 1 and 2 only · **Verdict:** Correct

The UNEP manual lists “immediate” objectives (better design, mitigation, informed decisions) and “long-term” objectives (sustainable development, avoiding irreversible damage). Pairs 3 and 4 misstate EIA practice.

Q149 · Sequence

According to the standard environmental waste management hierarchy, arrange the following waste management options in correct order from most preferred to least preferred

1. Disposal in a sanitary landfill
2. Recycling and composting of materials
3. Source reduction and waste minimization
4. Incineration with energy recovery

Which of the following represents the correct sequence?

Options:

1. 3 – 2 – 4 – 1
2. 2 – 3 – 4 – 1
3. 3 – 4 – 2 – 1
4. 1 – 2 – 3 – 4

Official key: (1) 3 – 2 – 4 – 1 · **Verdict:** Correct

The hierarchy runs from reduction, to reuse and recycling, to energy recovery, to disposal.

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Q150 · Sequence

Arrange the following milestones in chronological order:

1. Rio Declaration Principle 15 – Precautionary Approach
2. World Charter for Nature
3. Brundtland Commission Report
4. Wingspread Statement on the Precautionary Principle

Options:

1. 2 → 3 → 1 → 4
2. 3 → 2 → 4 → 1
3. 2 → 4 → 3 → 1
4. 4 → 2 → 3 → 1

Official key: (1) 2 → 3 → 1 → 4 · **Verdict:** Correct

The dates are the World Charter for Nature in 1982, the Brundtland Report in 1987, the Rio Declaration in 1992 and the Wingspread Statement in 1998.

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Wrong and disputed keys

The official key is correct for 146 of the 150 questions. Four can be challenged if the commission invites objections: Q86 and Q85 (wrong keys), Q25 (a flawed question) and Q127 (a weak case). Q86 is the strongest.

Q	Official key	Correct / better answer	Strength of objection
86	(4) I, II and III	(3) I and II only	Strong: the keyed answer is scientifically wrong
85	(4) I incorrect, II correct	(2) Both correct, II does not explain I	Moderate to strong: standard texts class detritus as abiotic
25	(3) D, B, A, C	Question flawed; seek deletion or bonus marks	Moderate: the premise contradicts Bernoulli's principle
127	(3) Dune stabilisation	(3) is the best answer; (2) Shelterbelts is arguable	Weak: do not rely on it

Two minor wording flaws do not change the answers. Q38 calls agricultural runoff an "air pollution" source. Q105 refers to "the context", a passage the paper does not print.

Q86: the key is wrong

Question: Which statements correctly describe trophic energy flow? I. Producers convert radiant energy into chemical energy. II. Food webs contain interconnected food chains. III. Energy is lost as decomposers recycle heat.

Official key: (4) I, II and III. **Correct answer:** (3) I and II only.

Detailed answer:

1. **Statement I is true.** Producers fix solar (radiant) energy as chemical energy in organic compounds through photosynthesis. Only about 2–10% of photosynthetically active radiation is captured (NCERT).
2. **Statement II is true.** A food web is a network of interconnected food chains.
3. **Statement III is false, for three reasons.**
    a. Heat is the degraded end-form of energy. By the second law of thermodynamics it cannot be re-converted into usable chemical energy by any organism, so it cannot be "recycled".
    b. Decomposers recycle nutrients (mineralisation), not energy. Their respiration releases the last of the energy in detritus as heat, which radiates out of the ecosystem.
    c. Ecology rests on the contrast that matter cycles while energy flows one way. A statement that energy (heat) is recycled reverses that rule.
4. Since III is false, only I and II are correct, which is option (3).

References:

- NCERT Biology Class XII, chapter "Ecosystem", section on energy flow. It states that energy flow is unidirectional, from the Sun to producers to consumers, and that energy is lost as heat at each level.
- NCERT Biology Class XII, chapter "Ecosystem", sections on decomposition and nutrient cycling. Decomposers break detritus into inorganic nutrients (mineralisation), and nutrients, not energy, are recycled.
- E.P. Odum & G.W. Barrett, *Fundamentals of Ecology* (5th ed.), chapter on energy in ecological systems. It covers the first and second laws applied to ecosystems and the one-way flow of energy, with degradation to heat.
- Erach Bharucha, *Textbook of Environmental Studies for Undergraduate Courses* (UGC), chapter on ecosystems, section on energy flow and nutrient cycling.

Q85: the key is very likely wrong

Question: Statement I: Dead leaf litter is an abiotic component even though it originated from plants. Statement II: Decomposer microorganisms convert organic matter into simpler substances available to producers.

Official key: (4) I incorrect, II correct. **Better answer:** (2) Both correct, but II does not explain I.

Detailed answer:

1. **Statement II is true** and undisputed. It is the definition of decomposition and mineralisation.
2. **Statement I is true under the standard textbook classification.** Ecosystems are divided into biotic (living) and abiotic (non-living) components. Odum's scheme lists the abiotic components as:
    a. Inorganic substances such as C, N, CO_2 and H_2O.
    b. **Organic compounds such as proteins, carbohydrates, lipids and humic substances.** These link the biotic and abiotic parts.
    c. The climate regime.
3. Dead leaf litter is dead organic matter (detritus). It is no longer living, so it falls in the abiotic, non-living organic fraction. Its plant origin does not make it biotic, which is exactly the point the statement's "even though" clause anticipates.
4. **Statement II does not explain Statement I.** What decomposers do to litter is a separate fact from why litter counts as abiotic, so option (2) follows, not option (1).
5. The key implies the examiners treated litter as biotic because of where it came from. That contradicts the standard definition. At best the statement is ambiguous, which is itself grounds for review.

References:

- E.P. Odum & G.W. Barrett, *Fundamentals of Ecology* (5th ed.), chapter on the ecosystem, section on the components of an ecosystem. It lists abiotic components including organic compounds and humic substances, and treats dead organic matter as non-living.
- Erach Bharucha, *Textbook of Environmental Studies* (UGC), chapter on ecosystems, section on the structure of an ecosystem (abiotic components, including organic matter).
- NCERT Biology Class XII, chapter "Ecosystem", section on decomposition. It defines detritus as dead plant remains such as leaves, bark and flowers, plus dead animal remains, which is the raw material for decomposition.

Q25: the question is flawed

Question: Four points on a streamline have heads A (20 m), B (25 m), C (15 m) and D (30 m). Arrange them in order of decreasing total head as per Bernoulli's principle.

Official key: (3) D, B, A, C. **Position:** the arithmetic is right, but the premise is invalid.

Detailed answer:

1. Bernoulli's equation along a streamline, for steady, incompressible, frictionless flow, is:

$$\frac{p}{\rho g} + \frac{V^2}{2g} + z = H = \text{constant}$$

2. So, "as per Bernoulli's principle", all four points on one streamline must have the **same** total head. With real-fluid losses, H can only **decrease** in the direction of flow.
3. Four different total heads (15, 20, 25 and 30 m) at arbitrary points cannot be ordered "as per Bernoulli's principle". The question is really just ranking four numbers.
4. In the exam, (3) is the only option that ranks them in descending order. For an objection, ask for the question to be deleted or bonus marks awarded, because the stem contradicts the principle it cites.

References:

- R.K. Bansal, *Fluid Mechanics and Hydraulic Machines*, chapter on dynamics of fluid flow (Bernoulli's equation, its assumptions, and the equation for real fluids with a loss term).
- F.M. White, *Fluid Mechanics*, chapter on the Bernoulli equation (Constant total head along a streamline in frictionless flow).

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Q127: the key holds, but there is a weak alternative

Official key: (3) Dune stabilisation. **Possible alternative:** (2) Shelterbelts.

Detailed answer: The stem describes moving sand dunes. Halting moving dunes requires fixing the sand in place with checkerboard micro-windbreaks, mulching and sand-binding vegetation. That is dune stabilisation, CAZRI's standard prescription for the Thar. Shelterbelts are rows of trees that reduce wind speed over fields and settlements downwind. They complement dune stabilisation but do not fix an active dune. The key is the most specific correct answer, so an objection is unlikely to succeed.

References: Erach Bharucha, *Textbook of Environmental Studies*, chapter on natural resources (Land degradation and desertification); CAZRI (Central Arid Zone Research Institute), Jodhpur, publications on sand-dune stabilisation and shelterbelt technology.

Standard references used throughout

- **Materials:** W.D. Callister, *Materials Science and Engineering*; R.K. Rajput, *Material Science and Engineering* and *Strength of Materials*; S.K. Duggal, *Building Materials*.
- **Fluids:** R.K. Bansal, *Fluid Mechanics and Hydraulic Machines*.
- **Water and wastewater:** IS 10500:2012; S.K. Garg, *Water Supply Engineering* and *Sewage Disposal and Air Pollution Engineering*; Metcalf & Eddy, *Wastewater Engineering*; CPHEEO manuals.
- **Air and noise:** CPCB NAAQS notification, 18 November 2009; Rao & Rao, *Air Pollution*; Peavy, Rowe & Tchobanoglous, *Environmental Engineering*.
- **Solid waste:** Tchobanoglous, Theisen & Vigil, *Integrated Solid Waste Management*; CPHEEO *MSWM Manual* (2016).
- **Thermodynamics and process calculations:** Smith, Van Ness & Abbott; D.M. Himmelblau.
- **Ecology, resources and biodiversity:** NCERT Biology XII; NCERT X Geography, *Contemporary India II*; Odum & Barrett, *Fundamentals of Ecology*; Erach Bharucha, *Textbook of Environmental Studies* (UGC).
- **Law:** Environment (Protection) Act, 1986; Wild Life (Protection) Act, 1972; NGT Act, 2010; SWM, BMW, Plastic and E-Waste Management Rules, 2016; *Vellore Citizens' Welfare Forum v. Union of India* (1996).
- **EIA:** EIA Notification, 2006; UNEP *EIA Training Resource Manual* (2002); ISO 14001 and ISO 19011.

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